

P146:

T₁: 由已知: 得 $\varepsilon = \sum_{i=1}^n (y_i - ax_i^2 - bx_i - c)^2$, 为使总偏差 $\varepsilon(a, b, c)$.

达到最小, 由极值必要条件, 有:

$$\frac{\partial \varepsilon}{\partial a} = -2 \sum_{i=1}^n (y_i - ax_i^2 - bx_i - c) x_i^2 = 0$$

$$\frac{\partial \varepsilon}{\partial b} = -2 \sum_{i=1}^n (y_i - ax_i^2 - bx_i - c) x_i = 0$$

$$\frac{\partial \varepsilon}{\partial c} = -2 \sum_{i=1}^n (y_i - ax_i^2 - bx_i - c) = 0$$

即 a, b, c 满足如下方程组:

$$\begin{cases} \left(\sum_{i=1}^n x_i^4 \right) a + \left(\sum_{i=1}^n x_i^3 \right) b + \left(\sum_{i=1}^n x_i^2 \right) c = \sum_{i=1}^n x_i^2 y_i \\ \left(\sum_{i=1}^n x_i^3 \right) a + \left(\sum_{i=1}^n x_i^2 \right) b + \left(\sum_{i=1}^n x_i \right) c = \sum_{i=1}^n x_i y_i \\ \left(\sum_{i=1}^n x_i^2 \right) a + \left(\sum_{i=1}^n x_i \right) b + nc = \sum_{i=1}^n y_i \end{cases}$$

T_2 :

Sol:

$$J(a, b) = \int_0^1 (y - \eta)^2 dx = \int_0^1 (x^2 - ax - b)^2 dx = \frac{1}{5} + \frac{a^2}{3} + b^2 - \frac{a}{2} - \frac{2}{3}b + ab.$$

为选择 a, b 使得 $J(a, b)$ 达到极小.

可得:

$$\frac{\partial J}{\partial a} = -\frac{1}{2} + \frac{2}{3}a + b = 0.$$

$$\frac{\partial J}{\partial b} = -\frac{2}{3} + a + 2b = 0.$$

$$\text{则 } a = 1, \quad b = -\frac{1}{6}.$$

于是曲线 $y = x^2$ 用直线 $\eta = x - \frac{1}{6}$ 来代替.

P151:

T₁:

(1).

解: 作函数 $L = x + y + \lambda(x^2 + y^2 - 1)$.

解方程组:

$$\begin{cases} L_x = 1 + 2\lambda x = 0 \\ L_y = 1 + 2\lambda y = 0 \\ L_\lambda = x^2 + y^2 - 1 = 0 \end{cases} \text{ 得: } \begin{cases} x_1 = \frac{\sqrt{2}}{2} \\ y_1 = \frac{\sqrt{2}}{2} \\ \lambda_1 = -\frac{\sqrt{2}}{2} \end{cases} \quad \begin{cases} x_2 = -\frac{\sqrt{2}}{2} \\ y_2 = -\frac{\sqrt{2}}{2} \\ \lambda_2 = \frac{\sqrt{2}}{2} \end{cases}$$

$$\text{又 } L_{xx} = 2\lambda \quad L_{xy} = 0 \quad L_{yy} = 2\lambda.$$

则

$$d^2L \circ \left(\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2}\right) = -\sqrt{2}(dx^2 + dy^2) < 0.$$

于是函数在 $(\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2})$ 处取到最高值 $\sqrt{2}$.

同理, 函数在 $(-\frac{\sqrt{2}}{2}, -\frac{\sqrt{2}}{2})$ 处取到极小值 $-\sqrt{2}$.

(2) 作函数 $L = x - 2y + 2z + \lambda(x^2 + y^2 + z^2 - 1)$.

解方程组

$$\begin{cases} L_x = 1 + 2\lambda x = 0 \\ L_y = -2 + 2\lambda y = 0 \\ L_z = 2 + 2\lambda z = 0 \\ L_\lambda = x^2 + y^2 + z^2 - 1 = 0 \end{cases} \text{ 得: } \begin{cases} x_1 = \frac{1}{3} \\ y_1 = -\frac{2}{3} \\ z_1 = \frac{2}{3} \\ \lambda_1 = -\frac{3}{2} \end{cases} \quad \begin{cases} x_2 = -\frac{1}{3} \\ y_2 = \frac{2}{3} \\ z_2 = -\frac{2}{3} \\ \lambda_2 = \frac{3}{2} \end{cases}$$

$$\text{又 } L_{x^2} = L_{y^2} = L_{z^2} = 2\lambda, \quad L_{xy} = L_{xz} = L_{yz} = 0.$$

则:

$$d^2L(x_0, y_0, z_0) = 3(dx^2 + dy^2 + dz^2) > 0.$$

于是函数在 $(-\frac{1}{3}, \frac{2}{3}, -\frac{2}{3})$ 处取到最小值 -3 .

同理可得:

函数在 $(\frac{1}{3}, -\frac{2}{3}, \frac{2}{3})$ 处取得最大值 3 .

13). 作 $L = xyz + \lambda(\frac{1}{x} + \frac{1}{y} + \frac{1}{z} - \frac{1}{a})$

解方程组

$$\begin{cases} L_x = yz - \frac{\lambda}{x^2} = 0 \\ L_y = xz - \frac{\lambda}{y^2} = 0 \\ L_z = xy - \frac{\lambda}{z^2} = 0 \\ L_\lambda = \frac{1}{x} + \frac{1}{y} + \frac{1}{z} - \frac{1}{a} = 0 \end{cases}$$

得: $x = y = z = 3a, \lambda = 8/a^4$.

$$\text{又 } L_{x^2}(3a, 3a, 3a) = L_{y^2}(3a, 3a, 3a) = L_{z^2}(3a, 3a, 3a) = 6a.$$

$$L_{xy}(3a, 3a, 3a) = \cancel{L_{xy}} L_{xz}(3a, 3a, 3a) = L_{yz}(3a, 3a, 3a) = 3a.$$

$$\text{则 } d^2L(3a, 3a, 3a) = 3a[(dx + dy + dz)^2 + dx^2 + dy^2 + dz^2] > 0.$$

于是函数在 $(3a, 3a, 3a)$ 处取到极小值 $27a^3$.

(5) 作函数 $L = xyz + u(x^2 + y^2 + z^2 - 1) + v(x + y + z)$

解方程组

$$L_x = yz + 2ux + v = 0$$

$$L_y = xz + 2uy + v = 0$$

得:

$$L_z = xy + 2uz + v = 0$$

$$L_u = x^2 + y^2 + z^2 - 1 = 0$$

$$L_v = x + y + z = 0$$

$$\left\{ \begin{array}{l} x_1 = \frac{\sqrt{6}}{6} \\ y_1 = \frac{\sqrt{6}}{6} \\ z_1 = -\frac{\sqrt{6}}{3} \\ u_1 = \frac{\sqrt{6}}{12} \\ v_1 = \frac{1}{6} \end{array} \right.$$

或

$$\left\{ \begin{array}{l} x_2 = -\frac{\sqrt{6}}{6} \\ y_2 = -\frac{\sqrt{6}}{6} \\ z_2 = \frac{\sqrt{6}}{3} \\ u_2 = -\frac{\sqrt{6}}{12} \\ v_2 = \frac{1}{6} \end{array} \right.$$

或

$$\left\{ \begin{array}{l} x_3 = -\frac{\sqrt{6}}{3} \\ y_3 = \frac{\sqrt{6}}{6} \\ z_3 = \frac{\sqrt{6}}{6} \\ u_3 = \frac{\sqrt{6}}{12} \\ v_3 = \frac{1}{6} \end{array} \right.$$

或

$$\left\{ \begin{array}{l} x_4 = \frac{\sqrt{6}}{3} \\ y_4 = -\frac{\sqrt{6}}{6} \\ z_4 = -\frac{\sqrt{6}}{6} \\ u_4 = -\frac{\sqrt{6}}{12} \\ v_4 = \frac{1}{6} \end{array} \right.$$

或

$$\left\{ \begin{array}{l} x_5 = \frac{\sqrt{6}}{6} \\ y_5 = -\frac{\sqrt{6}}{3} \\ z_5 = \frac{\sqrt{6}}{6} \\ u_5 = \frac{\sqrt{6}}{12} \\ v_5 = \frac{1}{6} \end{array} \right.$$

或

$$\left\{ \begin{array}{l} x_6 = -\frac{\sqrt{6}}{6} \\ y_6 = \frac{\sqrt{6}}{3} \\ z_6 = -\frac{\sqrt{6}}{6} \\ u_6 = -\frac{\sqrt{6}}{12} \\ v_6 = \frac{1}{6} \end{array} \right.$$

$$\text{又 } d^2L = 2u(dx^2 + dy^2 + dz^2) + 2(zdxdy + ydxdz + xdydz) \\ \text{则在点 } (x_1, y_1, z_1) \text{ 处, } d^2L = \frac{\sqrt{6}}{6}(dx^2 + dy^2 + dz^2 - 4dxdy + 2dxdz + 2dydz)$$

$$\text{又由 } x+y+z=0, \quad dx+dy+dz=0.$$

$$\text{则: } dx = -dy - dz = 0.$$

$$\Rightarrow d^2L(\text{在 } x_1, y_1, z_1) = \sqrt{6} dx^2 > 0$$

$$\text{则函数在 } (\frac{\sqrt{6}}{6}, \frac{\sqrt{6}}{6}, -\frac{\sqrt{6}}{3}) \text{ 处取到极小值 } -\frac{\sqrt{6}}{18}.$$

$$\text{同理可得, 函数在 } (x_3, y_3, z_3), (x_5, y_5, z_5) \text{ 处取到极小值 } -\frac{\sqrt{6}}{18}.$$

$$\text{在 } (x_1, y_1, z_1), (x_4, y_4, z_4), (x_6, y_6, z_6) \text{ 处取到最大值 } \frac{\sqrt{6}}{18}.$$

2.

Sol: 因 $x>0, y>0, z>0$, 则 $f = x^m y^n z^p$ 最大时, $\ln f = m \ln x + n \ln y + p \ln z$ 也最大. 反之亦然, 故只需求 $\ln f$ 的最大值.

$$\text{令 } L = m \ln x + n \ln y + p \ln z + \lambda (x+y+z-a).$$

$$\text{则解方程 } \left\{ \begin{array}{l} L_x = \frac{m}{x} + \lambda = 0 \\ L_y = \frac{n}{y} + \lambda = 0 \\ L_z = \frac{p}{z} + \lambda = 0 \\ L_\lambda = x+y+z-a=0 \end{array} \right. \Rightarrow \left\{ \begin{array}{l} x = \frac{ma}{m+n+p} \\ y = \frac{na}{m+n+p} \\ z = \frac{pa}{m+n+p} \\ \lambda = -\frac{m+n+p}{a} \end{array} \right.$$

故在 $(\frac{ma}{m+n+p}, \frac{na}{m+n+p}, \frac{pa}{m+n+p})$ 处 ~~有~~ 为可能极值点

$$\text{又 } L_x^2 = -\frac{m}{x^2}, \quad L_y^2 = -\frac{n}{y^2}, \quad L_{xy} = L_{yz} = L_{xz} = 0.$$

$$L_z^2 = -\frac{p}{z^2}, \quad dL^2 = (-\frac{m}{x^2} dx^2 - \frac{n}{y^2} dy^2 - \frac{p}{z^2} dz^2) < 0$$

故在 $(\frac{ma}{m+n+p}, \frac{na}{m+n+p}, \frac{pa}{m+n+p})$ 处 $\ln f$ 有极大值.

$$\text{即 } f \text{ 有极大值 } \frac{m^m n^n p^p}{(m+n+p)^{m+n+p}} \cdot a^{m+n+p}.$$

又 $f = x^m n^n z^p$ 当 (x, y, z) 趋于边界 $\begin{cases} x+y=a, \\ z=0. \end{cases} \begin{cases} x+z=a, \\ y=0. \end{cases}$

$\begin{cases} y+z=a \\ x=0 \end{cases}$ 时, $f \rightarrow 0$. 故 f 的唯一极大点也是最大点.

例 3:

Sol: 设三角形另一顶点为 $(x, y) (x, y > 0)$

故问题可转换成求 $S = x(y+z)$ 在限制条件 $x^2 + 3y^2 = 12, x > 0$

$\max (x, y > 0)$.

$$\text{作函数 } L = x(y+z) + \lambda(x^2 + 3y^2 - 12).$$

极值点

解方程：
$$\begin{cases} L_x = y + 2 + 2\lambda x = 0 \\ L_y = x + 6\lambda y = 0 \\ L_z = x^2 + 3y^2 - 12 = 0 \end{cases} \quad \text{得：} \begin{cases} x = 3 \\ y = 1 \\ \lambda = -\frac{1}{6} \end{cases}$$

于是顶点坐标 $A(0, 2)$, $B(3, 1)$, $C(-3, -1)$ 或

$A(0, -2)$, $B(3, 1)$, $C(-3, 1)$.

故在 $(0, 2)$, $(3, -1)$ 及 $(-3, -1)$ 和

$(0, -2)$, $(3, 1)$ 及 $(-3, 1)$ 处其面积最大.

其值为 9.

T4:

解：设所求点坐标为 (x, y) ，则它到直线的距离为 $d = \frac{1}{\sqrt{2}} |x - y + 4|$.

其中 $y^2 = 4x$.

直线 $x - y + 4 = 0$ 将平面分成左右两部分，左面 $x - y + 4 < 0$.

右面 $x - y + 4 > 0$.

故点 (x, y) 到它的距离为

$$d = \frac{1}{\sqrt{2}} |x - y + 4|.$$

$$\text{令 } L = \frac{1}{\sqrt{2}} (x - y + 4) + \lambda (y^2 - 4x).$$

则解方程组:

$$\begin{cases} L_x = \frac{1}{\sqrt{2}} - 4\lambda = 0 \\ L_y = -\frac{1}{\sqrt{2}} + 2\lambda y = 0 \\ L_\lambda = y^2 - 4x = 0 \end{cases} \quad \text{得:} \quad \begin{cases} x=1 \\ y=2 \\ \lambda = \frac{1}{4\sqrt{2}} \end{cases}$$

又 $L_{xx} = 0, L_{yy} = \frac{1}{\sqrt{2}}, L_{xy} = 0, d^2L(1,2) = L_{xx}dx^2 + 2L_{xy}dxdy + L_{yy}dy^2 = \frac{dy^2}{\sqrt{2}} > 0$

故 (1,2) 为极小点. 即点 (1,2) 到直线的距离最近.

15:

解: 据题意: 求距离 $d = \sqrt{x^2 + y^2 + z^2}$. 在限制条件 $z = x^2 + y^2$, $x + y + z = 1$ 的最小值作 $L = x^2 + y^2 + z^2 + \lambda(z - x^2 - y^2) + \gamma(x + y + z - 1)$.

解:
$$\begin{cases} L_x = 2x - 2\lambda x + \gamma = 0 \\ L_y = 2y - 2\lambda y + \gamma = 0 \\ L_z = 2z + \lambda + \gamma = 0 \\ L_\lambda = z - x^2 - y^2 = 0 \\ L_\gamma = x + y + z - 1 = 0 \end{cases} \quad \text{得:} \quad \begin{cases} x_1 = \frac{-1+\sqrt{3}}{2} \\ y_1 = \frac{-1+\sqrt{3}}{2} \\ z_1 = 2-\sqrt{3} \\ \lambda_1 = \frac{-5\sqrt{3}+9}{3} \\ \gamma_1 = -7 + \frac{11}{3}\sqrt{3} \end{cases} \quad \text{或} \quad \begin{cases} x_2 = \frac{-1-\sqrt{3}}{2} \\ y_2 = \frac{-1-\sqrt{3}}{2} \\ z_2 = 2+\sqrt{3} \\ \lambda_2 = \frac{5\sqrt{3}+9}{3} \\ \gamma_2 = -7 - \frac{11}{3}\sqrt{3} \end{cases}$$

于是 $d(x_1, y_1, z_1) = \sqrt{9-5\sqrt{3}}$ $d(x_2, y_2, z_2) = \sqrt{9+5\sqrt{3}}$.

则最长距离为原点到点 $(-\frac{\sqrt{3}+1}{2}, -\frac{\sqrt{3}+1}{2}, 2+\sqrt{3})$ 的距离, 为

$$\sqrt{9+5\sqrt{3}}.$$

最短距离为原点到点 $(\frac{-1+\sqrt{3}}{2}, \frac{-1+\sqrt{3}}{2}, 2-\sqrt{3})$ 的距离, 为

$$\sqrt{9-5\sqrt{3}}.$$

T6:

解: 设 (x, y, z) 为平面 $Ax + By + Cz + D = 0$ 上任一点. 则它与 (a, b, c) 点的距离为 $d = \sqrt{(x-a)^2 + (y-b)^2 + (z-c)^2}$, 其中 (x, y, z) 满足

$$Ax + By + Cz + D = 0.$$

因 $d > 0$, 故 d 最大 $\Leftrightarrow d^2$ 最大.

按题设, 应求 $d^2 = (x-a)^2 + (y-b)^2 + (z-c)^2$ 在条件 $Ax + By + Cz + D = 0$ 下的极值:

$$\text{令 } L = (x-a)^2 + (y-b)^2 + (z-c)^2 + \lambda (Ax + By + Cz + D)$$

则解方程组

$$L_x = 2(x-a) + \lambda A = 0,$$

$$L_y = 2(y-b) + \lambda B = 0,$$

$$L_z = 2(z-c) + \lambda C = 0.$$

$$L_{\lambda} = Ax + By + Cz - D = 0$$

得:

$$x = a - \frac{1}{2}\lambda A.$$

$$y = b - \frac{1}{2}\lambda B.$$

$$z = c - \frac{1}{2}\lambda C.$$

$$\lambda = \frac{2(Aa + Bb + Cc + D)}{A^2 + B^2 + C^2}.$$

$$\text{于是 } d = \frac{|Aa + Bb + Cc + D|}{\sqrt{A^2 + B^2 + C^2}}.$$

当 x, y, z 任 $\rightarrow \infty$ 时, $d \rightarrow \infty$.

$\{(x, y, z) \mid (x-a)^2 + (y-b)^2 + (z-c)^2 < d\}$ 内必取最小值.

则点 (a, b, c) 到平面 $Ax + By + Cz + D = 0$ 的最短

$$\text{距离 } d = \frac{|Aa + Bb + Cc + D|}{\sqrt{A^2 + B^2 + C^2}}.$$